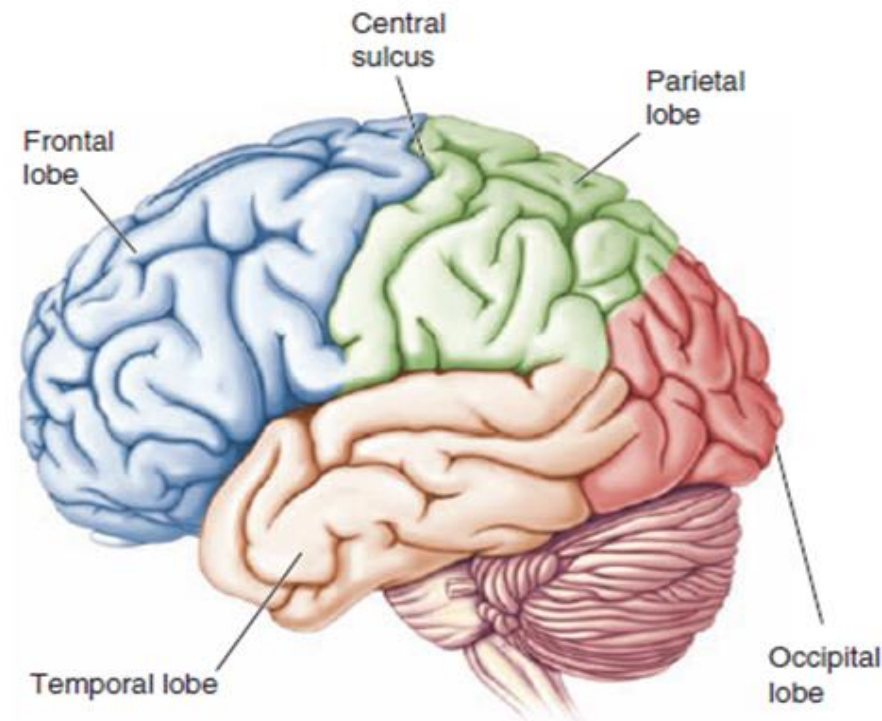

Central Nervous System

Cortical Motor Areas

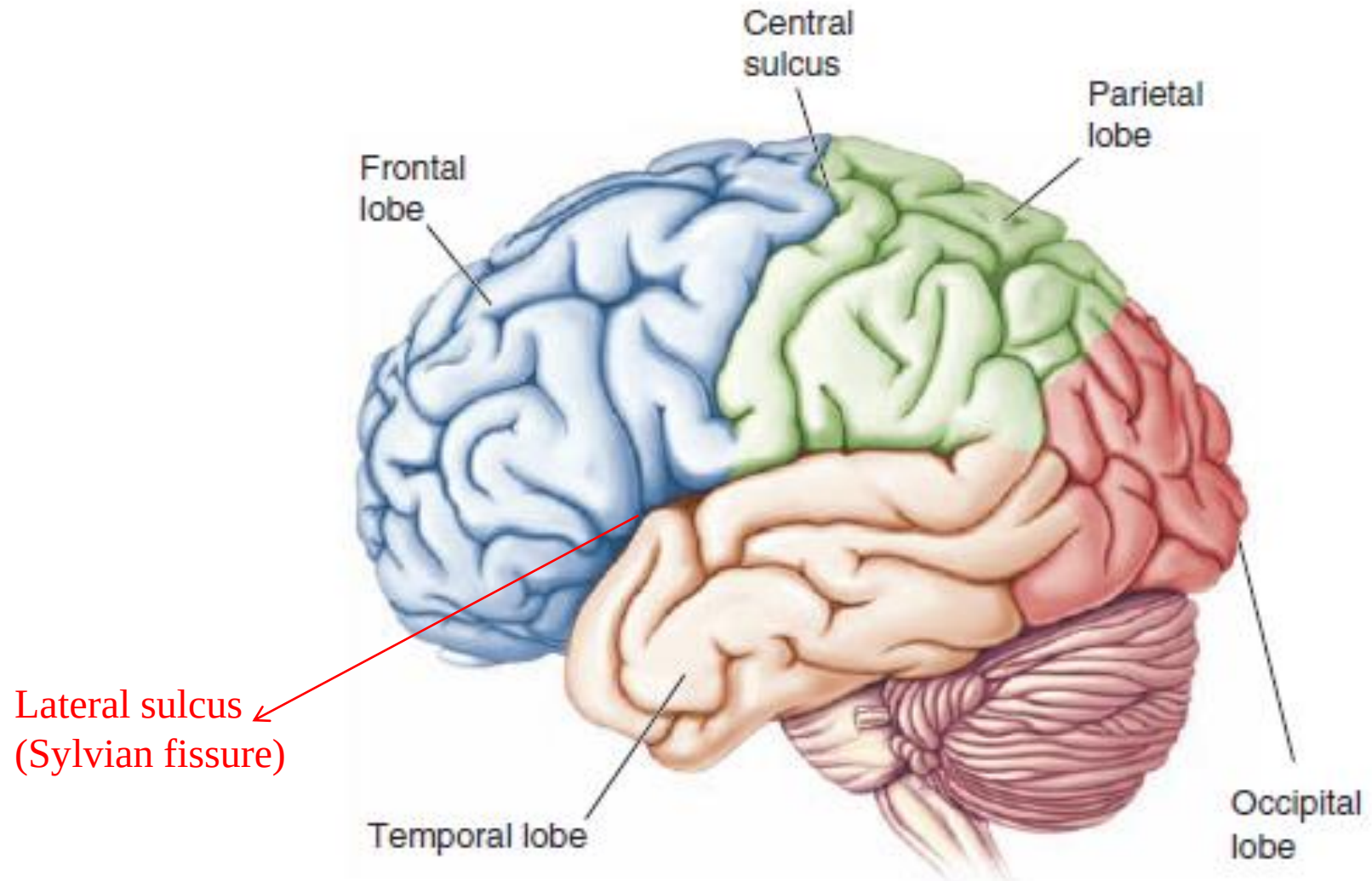
Cerebral Cortex

- Covers the entire surface of the brain (+ deep nuclei)
- Highly folded, forming gyri and sulci (↑SA, 80% of the mass of the brain)
- 2 hemispheres (rt. + lt.):
- Each cerebral hemisphere: 5 lobes:
 - Frontal lobe
 - Parietal lobe
 - Temporal lobe
 - Occipital lobe
 - Limbic lobe



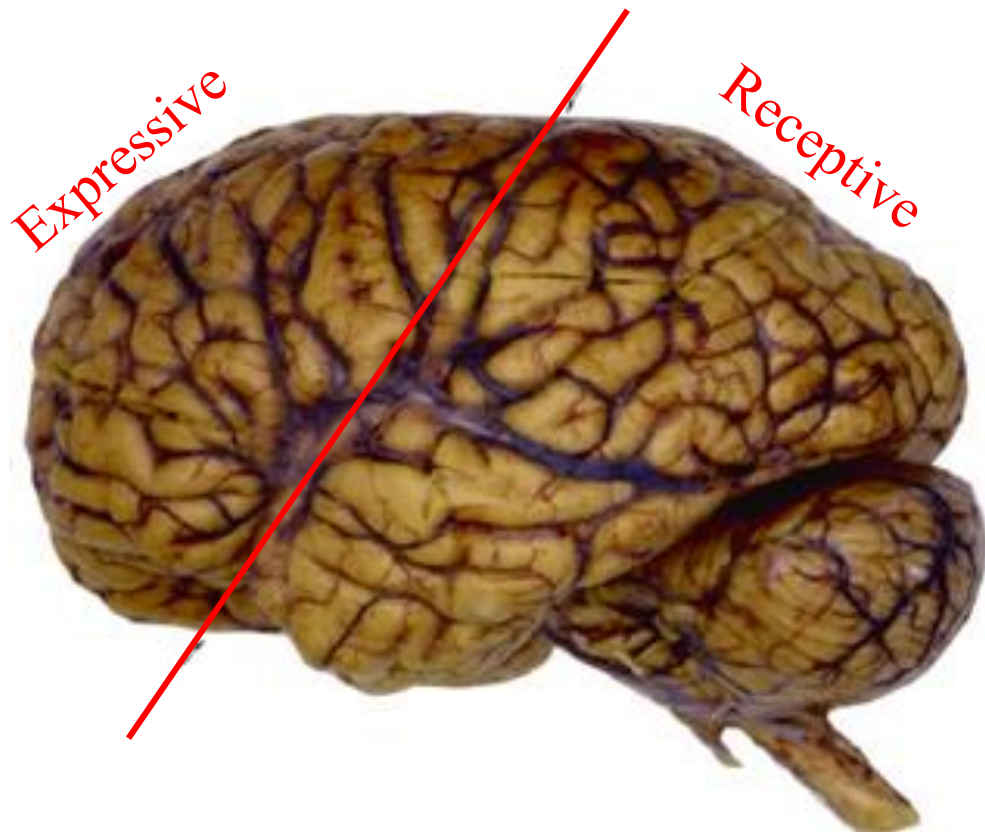
Cerebral Cortex

- Brain is organized into functionally specific areas...connected to form functional systems.



Cerebral Cortex

- The highest center of the nervous system
- Higher intellectual functions:
 - Speech + language
 - Planning + Judgment...
 - Intelligence
 - Learning + memory...
- Cerebral cortex...contralateral division of labor
 - Rt. Hemisphere → controls left side of the body and visual field
 - Lt hemisphere → controls right side of the body and visual field



Cerebral cortex

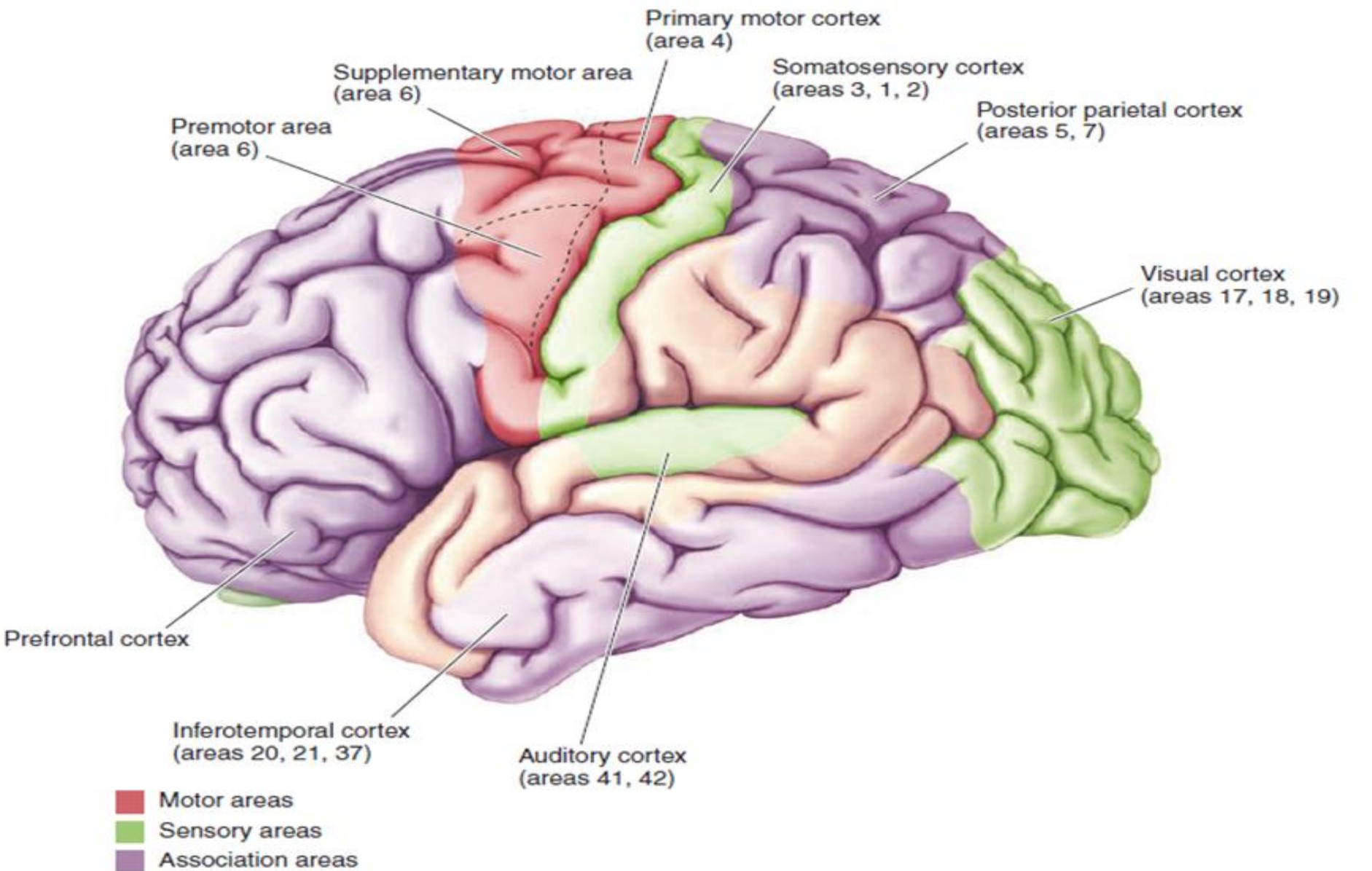
- ✓ CC is the site of the highest order integration of sensory, motor + consciousness activities.
- ✓ Cerebral dominance:
 - Dominant hemisphere (Lt.)
 - Language/symbolic thinking
 - Speech + writing
 - Calculation
 - Nondominant hemisphere (Rt.)
 - Spatial perception
 - Music/singing...
 - Playing musical instrument
 - Simple language comprehension

Cortical Motor Areas

- Cortical motor areas lie largely in Brodmann's area 4 + area 6
 1. Primary motor cortex/Area 4
 2. Premotor cortex/Area 6
 3. Supplementary motor cortex
 4. Motor association area → orbitofrontal + ventromedial prefrontal cortex

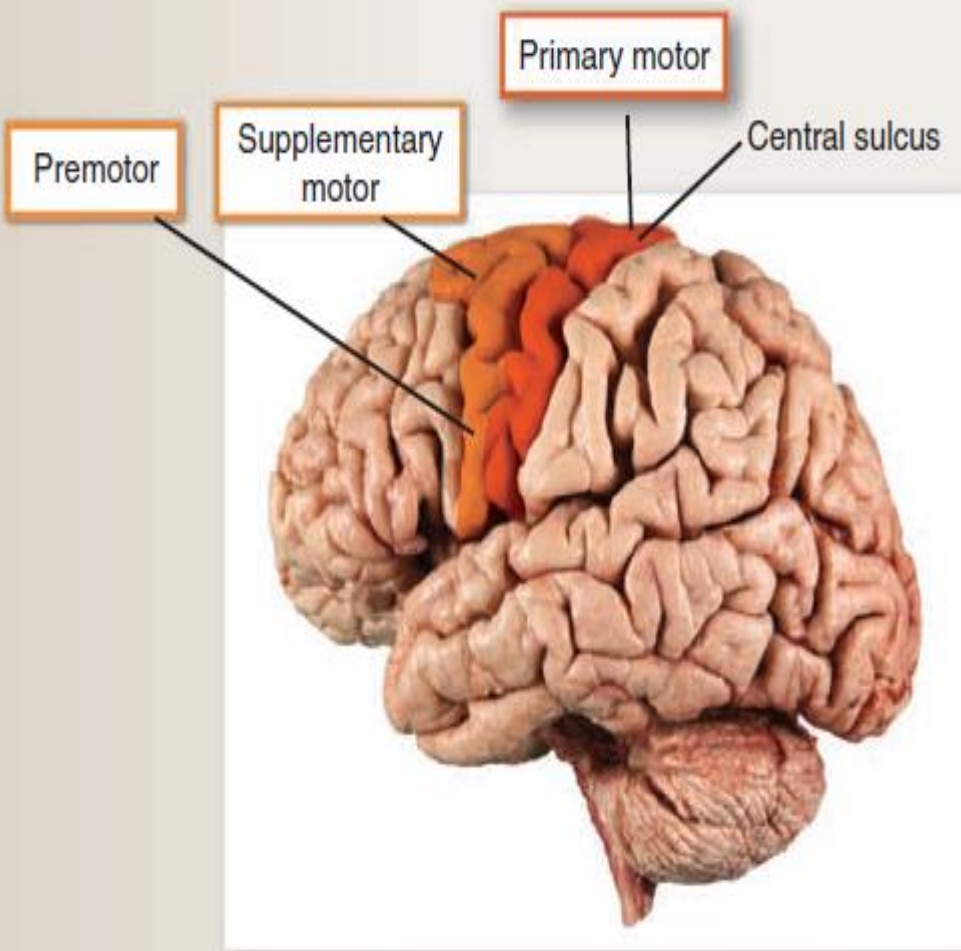
Primary Motor Cortex

- Localized in the precentral gyrus of the frontal lobes
 - Precentral gyrus of the left hemisphere sends motor output to the right side of the body and vice versa
- Output from the 1° motor cortex makes up the **corticospinal tract**
- Neurons in the 1° motor cortex are clustered in functional areas representing the various muscle groups they influence
- Receive direct + indirect input from association areas and their axons project to the **contralateral limb**

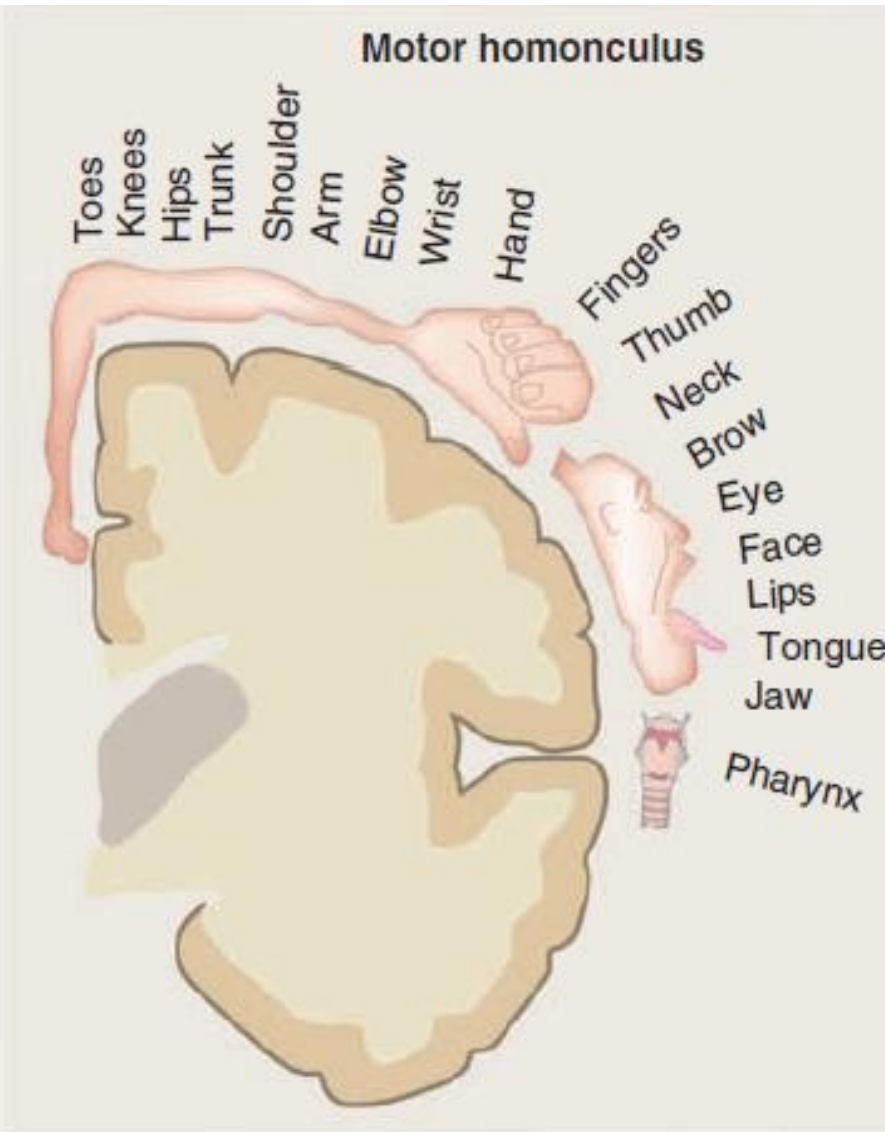


Primary Motor Cortex

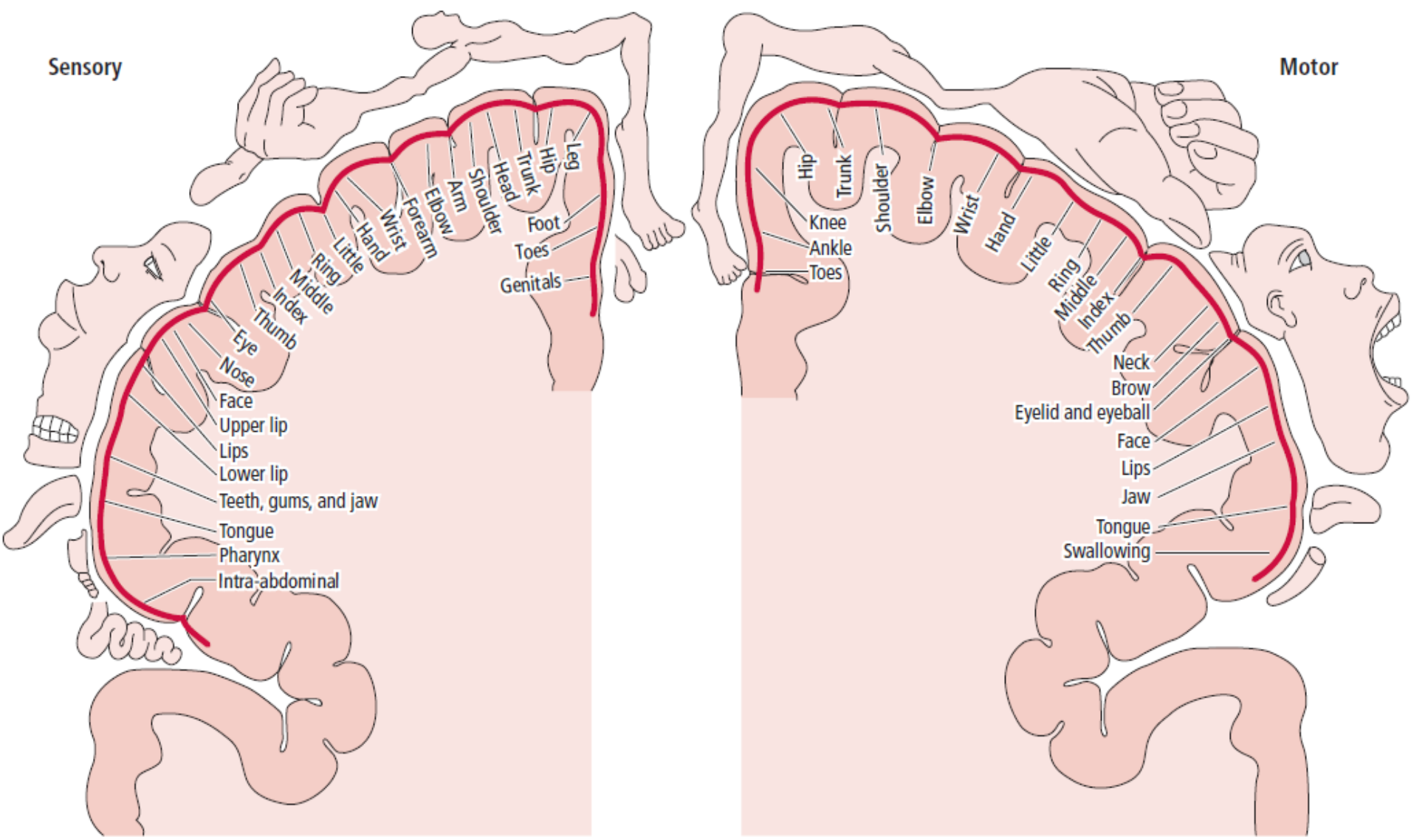
- Motor homunculus/Somatotopy:
 - The body is represented in an **inverted + crossed** manner
 - The size of body parts of the homunculus represents the size of the neuronal pool supplying the musculature of that part of the body
 - **Size of the area** of representation of each part of the body is proportionate to **the motor activity** and **degree of fine movements** of each part and not to its size:
 - *This reflects the larger number of neurons needed for fine motor control*
 - eg., hands and muscles of speech vs. trunk



Primary motor cortex

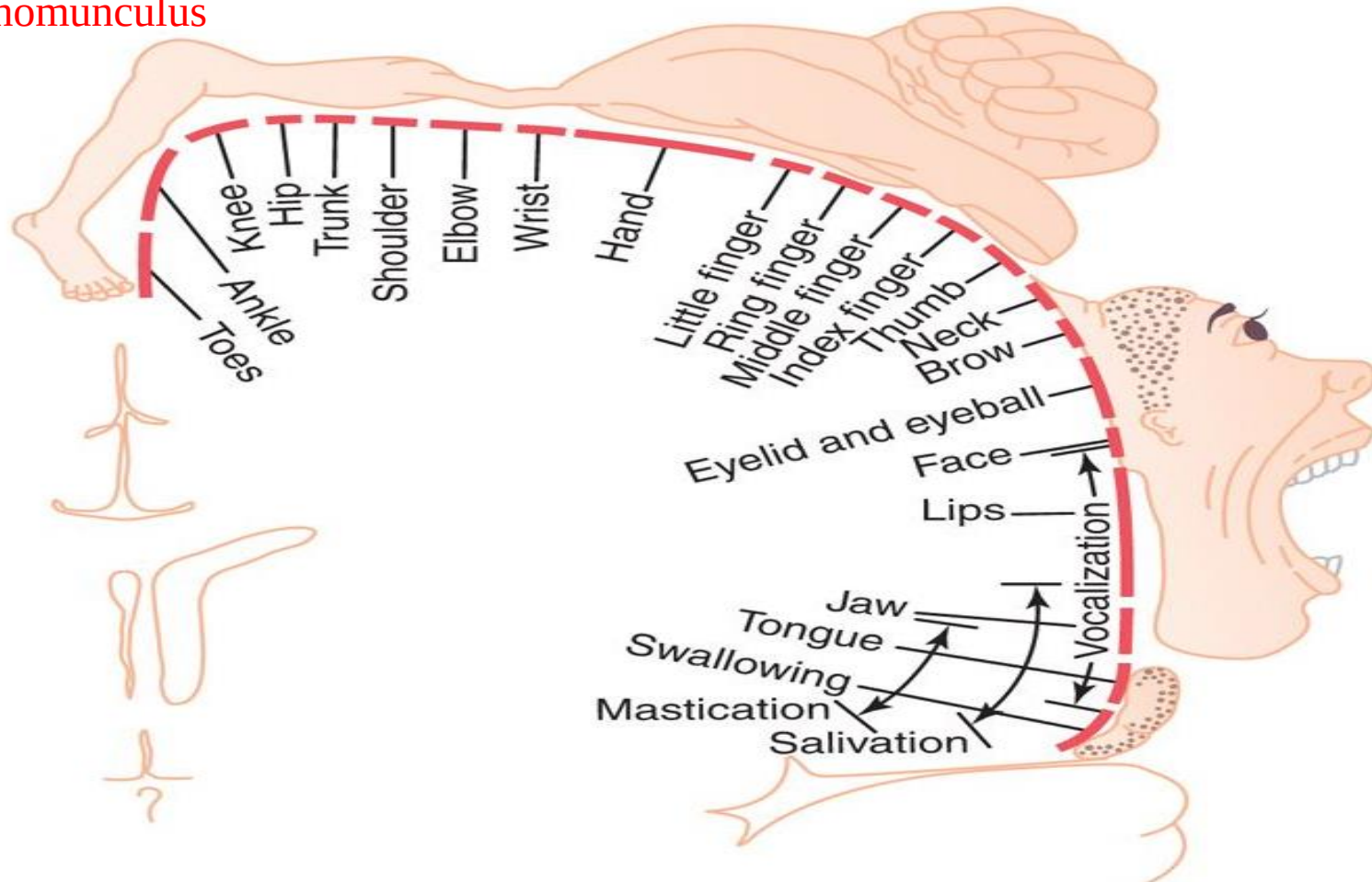


Motor homunculus



Sensory + motor homunculus

Motor homunculus



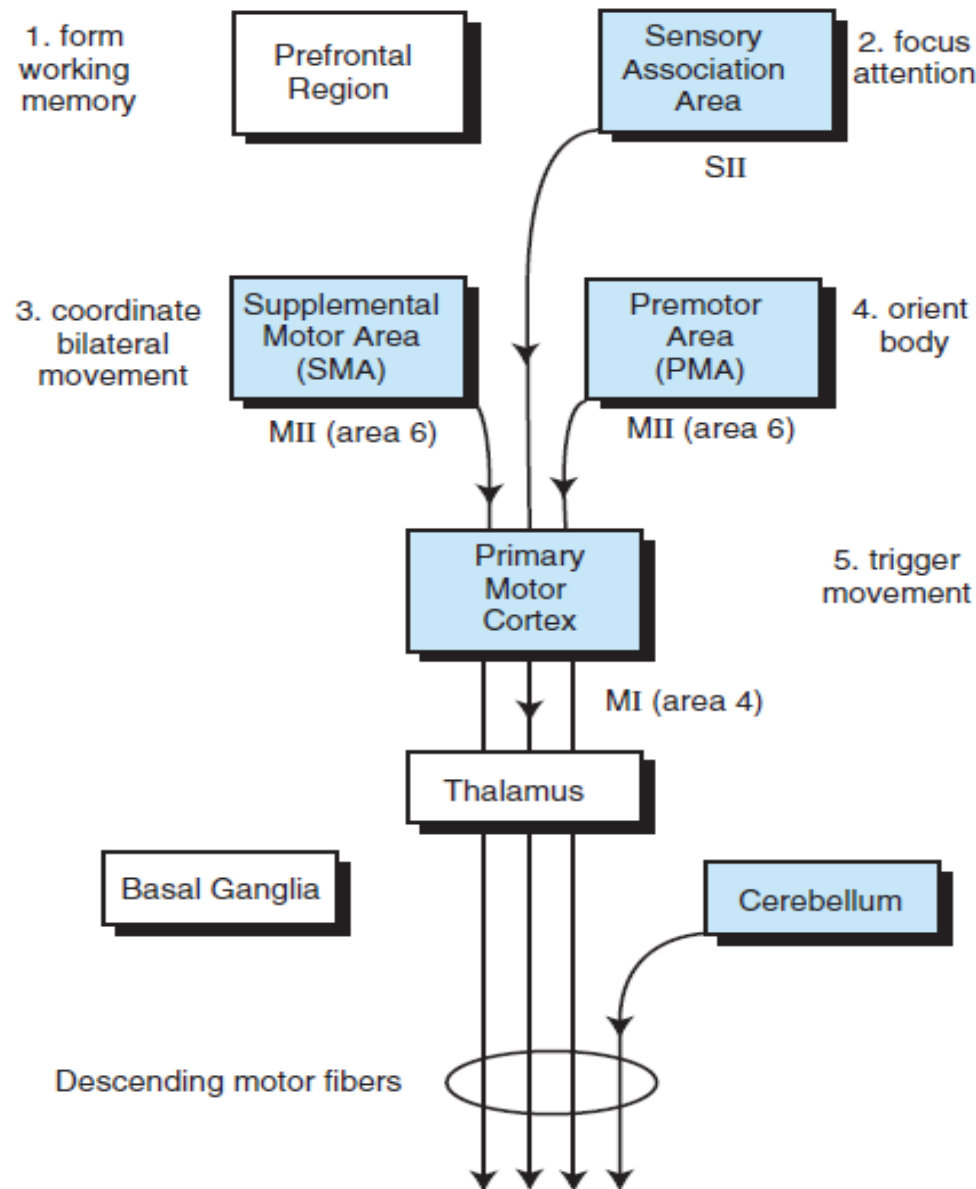
- ✓ The size of cortical surface dedicated to a part of the body is related to the degree of motor control of that part;
- ✓ Hence in humans much of the motor cortex is dedicated to moving the muscles of the fingers and the muscles related to speech.

Primary Motor Cortex

- **NB**
 - The motor map is **dynamic and adaptable**:
 - Continually shaped by an individual's motor experience
 - Functional reorganization often occurs after a focal lesion...recovery of function
 - Learning a motor skill changes organization of the motor map

Primary Motor Cortex

- Functions:
 - Initiation of voluntary, fine, discrete movements of the distal parts of the body (eg., hands + fingers)
 - Modulated by a continuous stream of information from other motor regions as well as tactile, visual, & proprioceptive information needed to make voluntary movement both accurate & properly sequenced
 - Facilitatory effect on stretch reflexes + skeletal muscle tone



- Voluntary movement requires coordination of all components of the motor system (motor cortex, basal ganglia, thalamus, midbrain, cerebellum and spinal cord ‘final common pathway’).

Premotor Area

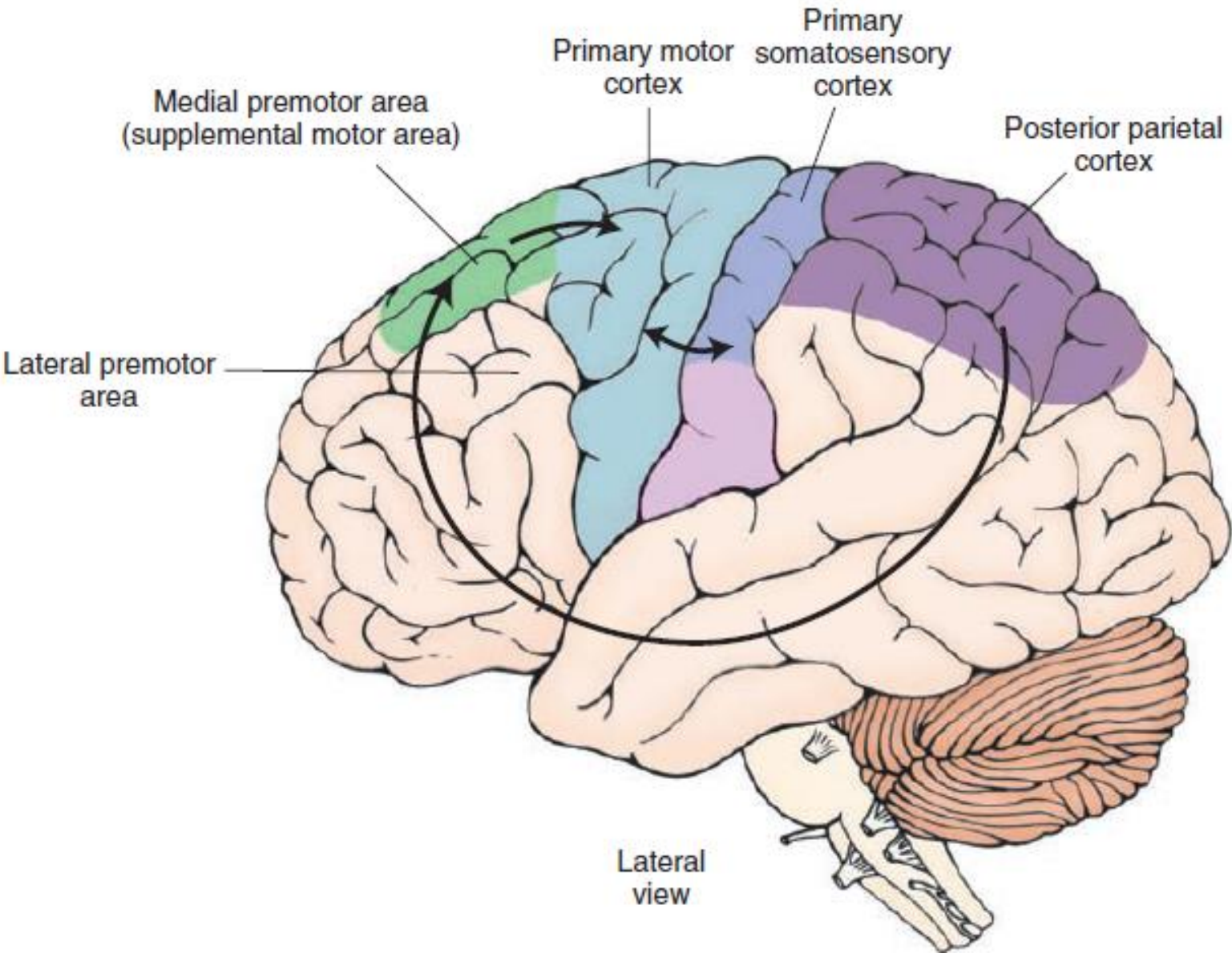
- Premotor cortex:
 - Lies immediately anterior to the PMC on the lateral hemisphere
 - Consists of several non-homogeneous distinct sub-areas which;
 - Have specific connections among themselves & with the rest of the cerebral cortex
 - Separately controls movements of some or all parts of the body
- Planning phase of motor activity
 - Control of axial and proximal limb musculature
 - Plays important role in movements that require visual guidance

Premotor Area

- **Supplementary motor area/SMA**
 - Located anterior to the PMC on the superior and medial aspects of the hemisphere
 - Topographic representation is **bilateral + horizontal** position (head → anteriorly, legs → posteriorly)/coarse map of the body
 - Contains motor maps for postures
 - Its efferents generally terminate bilaterally (**postural stabilization**)
 - *Programming sequential motor actions + coordination + postural adjustments*

Premotor Area

- Premotor areas are connected to the primary motor cortex (*somatotopically organized reciprocal connections + with each other also*)
- Have dense reciprocal connections with the association areas in the **posterior parietal cortex...**(*reciprocal, convergent, and divergent projections*)
 - Receive somatotopically organized input from S-I and parietal association cortex (*integrate somatic + visual sensations*)
 - These somatosensory and parietal inputs provide the primary motor cortex and caudal premotor regions with sensory information **to organize and guide motor acts**

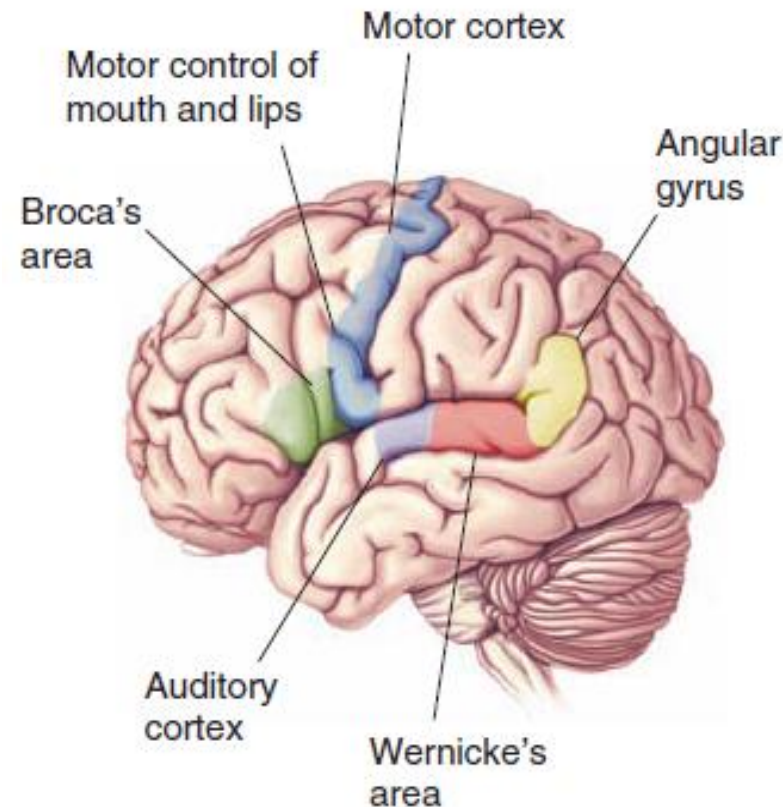


Premotor Area

- Some associated specialized areas with specific functions:
 - Broca's area → Area 44
 - Frontal eye field area → Area 8
 - Head rotation area → part of area 6
 - Hand skills area → part of area 6

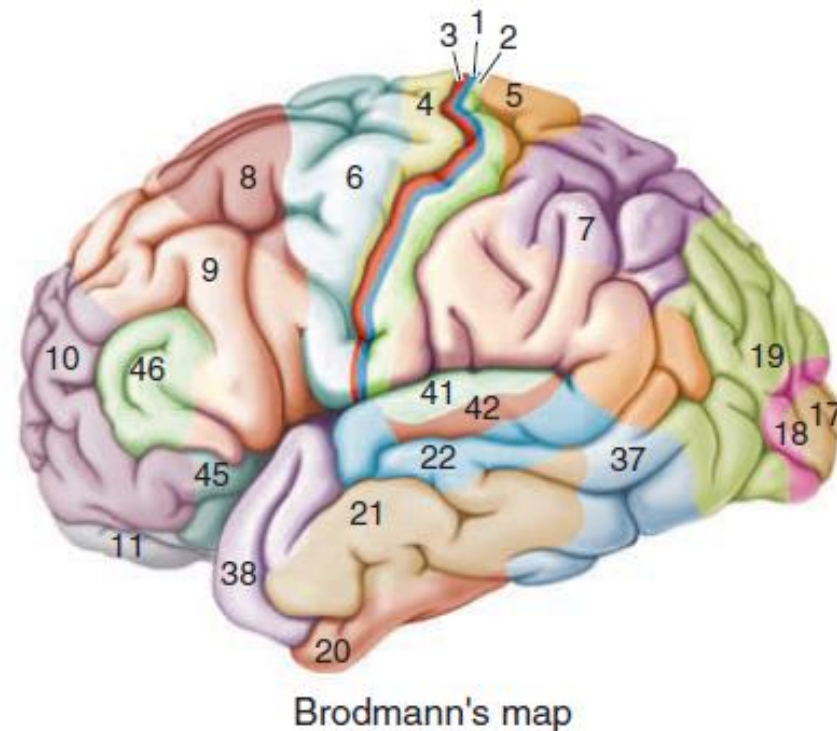
Premotor Area

- Broca's Area /Area 44
 - Located at the upper border of the lateral sulcus in front of the primary motor cortex
- Motor programs for verbalization/ dedicated to the production of language (spoken + written + sign language)
- Damage leads to inability to speak whole words/except as 'yes' or 'no'



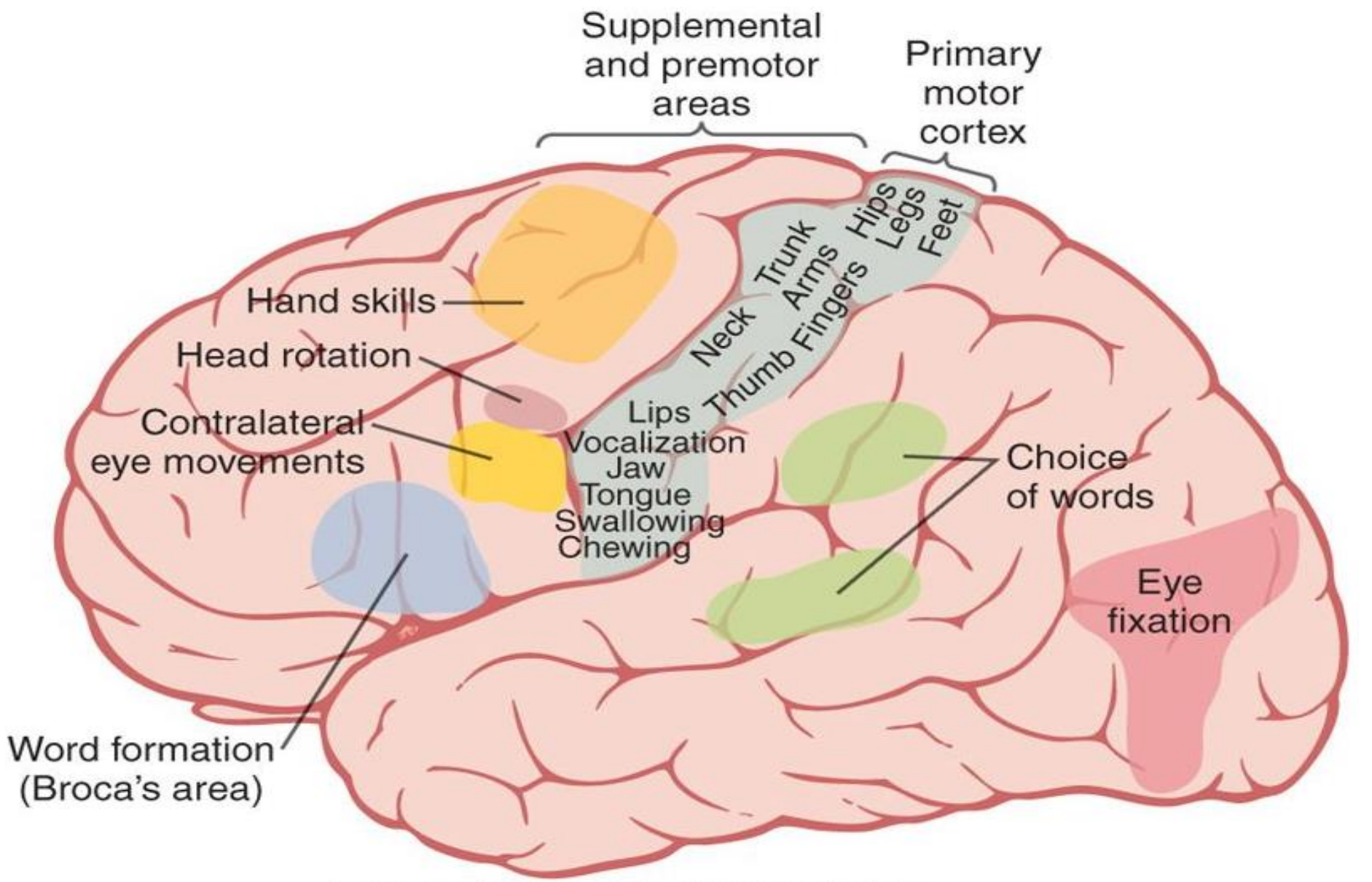
Premotor Area

- Frontal eye field area/Area 8
 - Lies above Broca's area
 - Visual tracking of a moving object
 - Control movements of the eyes and eyelids
 - Damage of this area leads to **locking/fixation** of the eye on objects



Premotor Area

- Head Rotation Area/part of area 6
 - Lies immediately above area 8 + Works in close association with it
 - Guiding/turning the body towards the desired directions
- Hand Skills Area/part of area 6
 - Lies anterior to the primary motor area for hands and fingers
 - Stores the motor programs for skilled hand movements
 - Damage of this area leads to 'Motor Apraxia' → inability to do skilled hand movements



Premotor Area

- Functions of Premotor Area:
 - Initiation of gross movements that involve groups of muscles to support + facilitate fine movement
 - Rotation of the head towards objects
 - Verbalization of words/area 44
 - Voluntary eye and eyelid movements/area 8
 - Skilled hand movement/ area 6

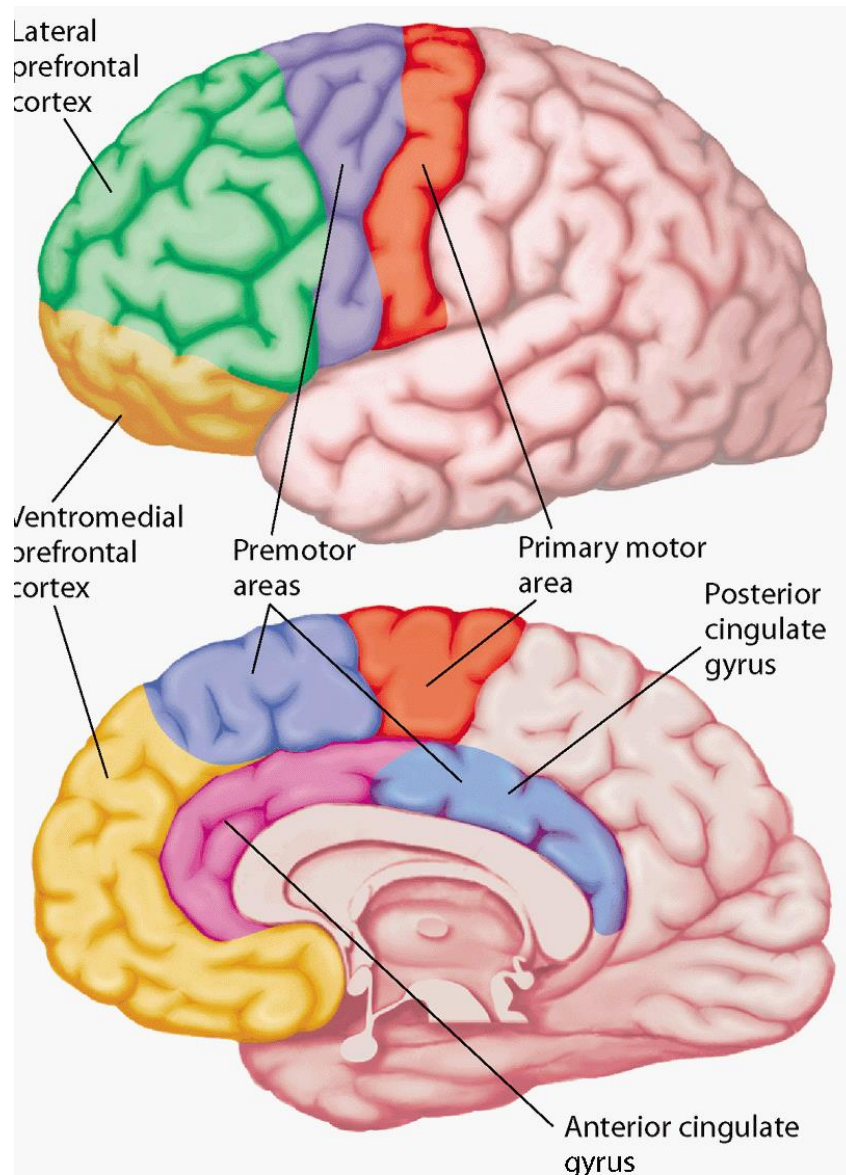
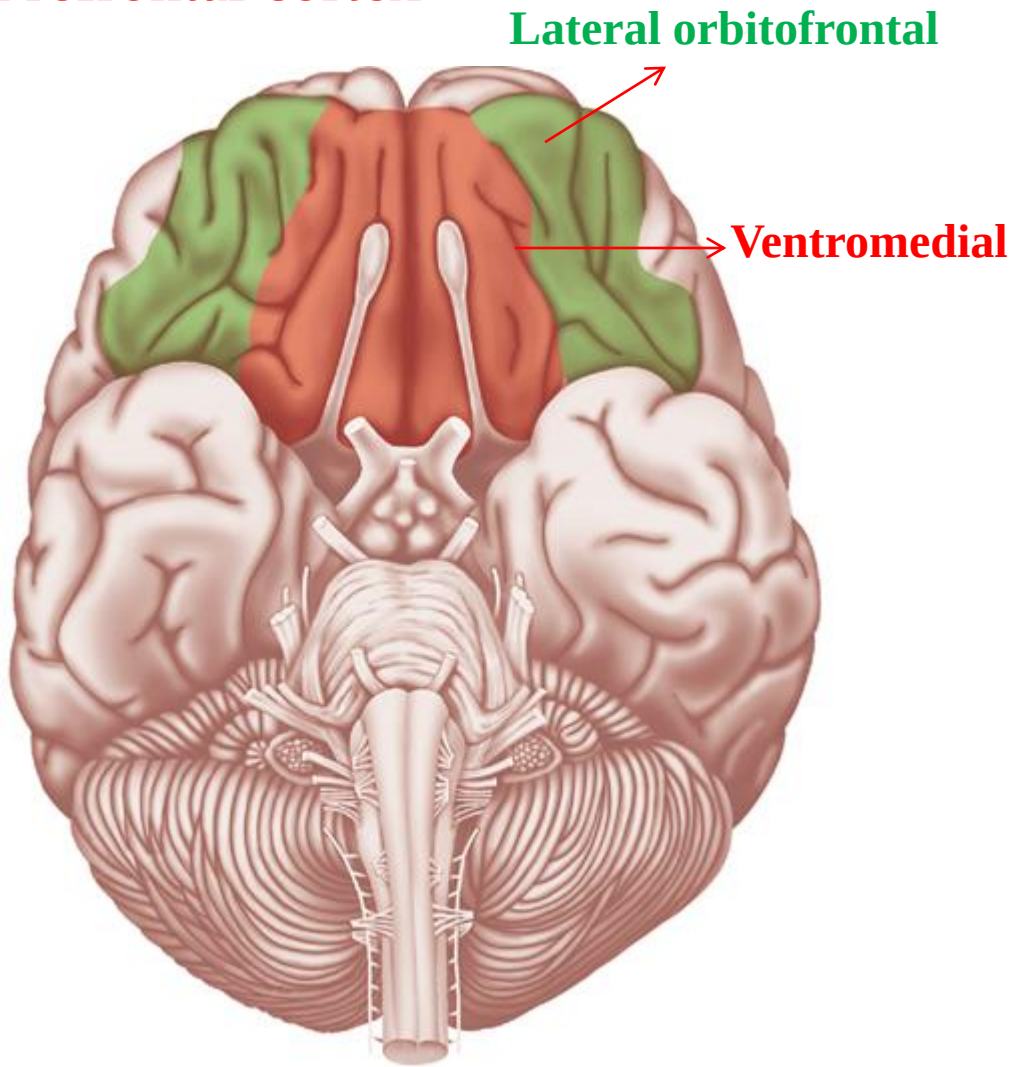
Premotor Area

- Acts on motor cells in the primary motor area in two ways:
 - Direct → projection fibers to area 4
 - Indirect → premotor area → corpus striatum/basal ganglia → thalamus → area 4 of the cerebral cortex
- Each premotor area has neurons that project to the brainstem as well as neurons that project directly to the spinal cord

Motor Association Area

- Frontal association area/Prefrontal cortex
 - Located in the frontal lobe just anterior to the premotor area (*spanning middle + inferior frontal gyri on the lateral side of the cortex*)
 - Extensively interconnected with other brain structures
 - Divided into two main regions (connectivity + function):
 1. Dorsolateral prefrontal cortex /Dorsal/superior + lateral parts/
 2. Orbitomedial prefrontal cortex /Ventral/inferior + medial portions/

Prefrontal cortex

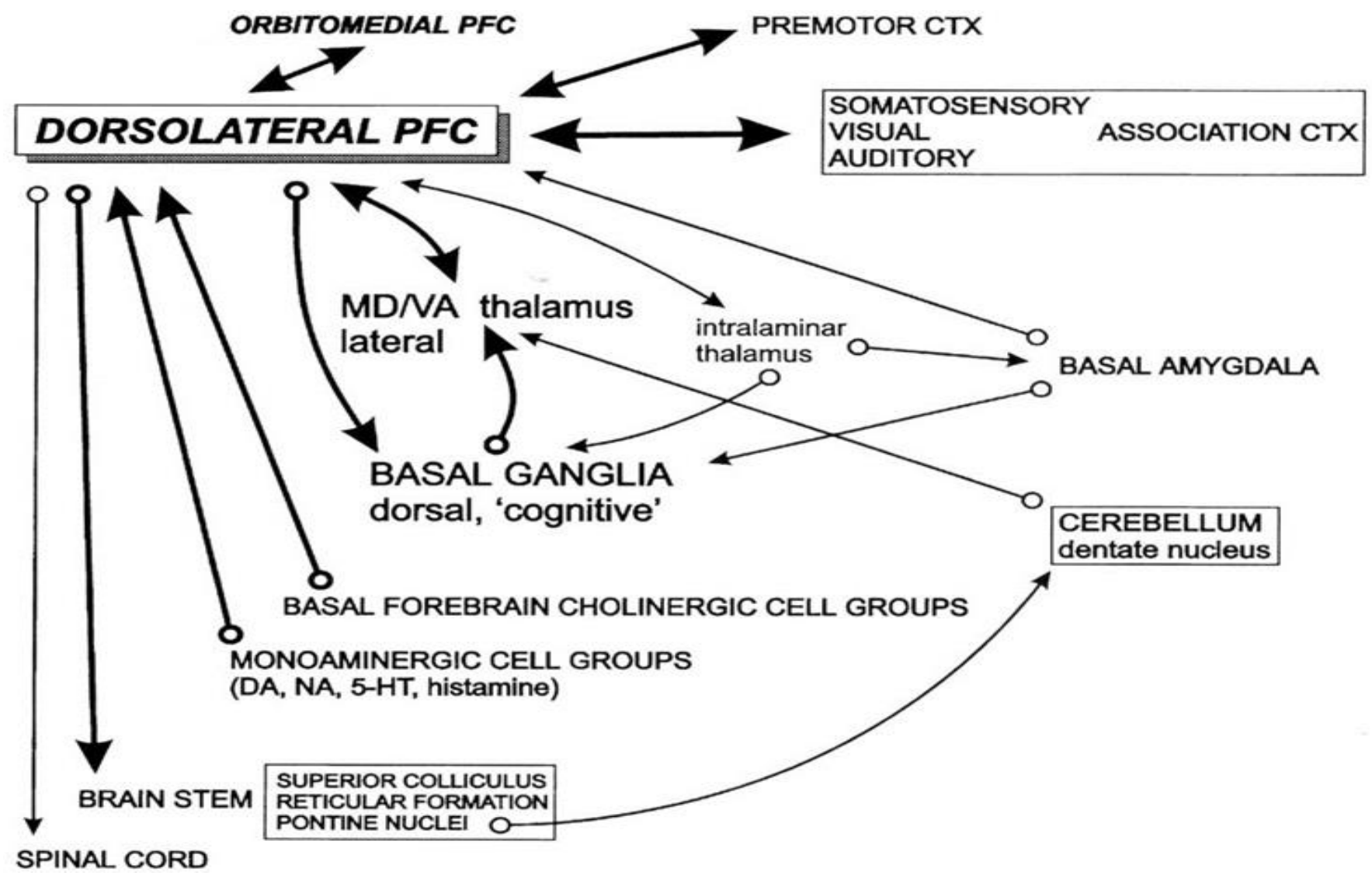


Motor Association Area

1. Dorsolateral prefrontal cortex

- Connected with sensory + motor cortex + basal ganglia + cerebellum + ...
- Regulate attention + motor responses to stimuli.../*maintaining intention* (disorganized behavior & distractibility)

Connections of Dorsolateral Prefrontal Cortex

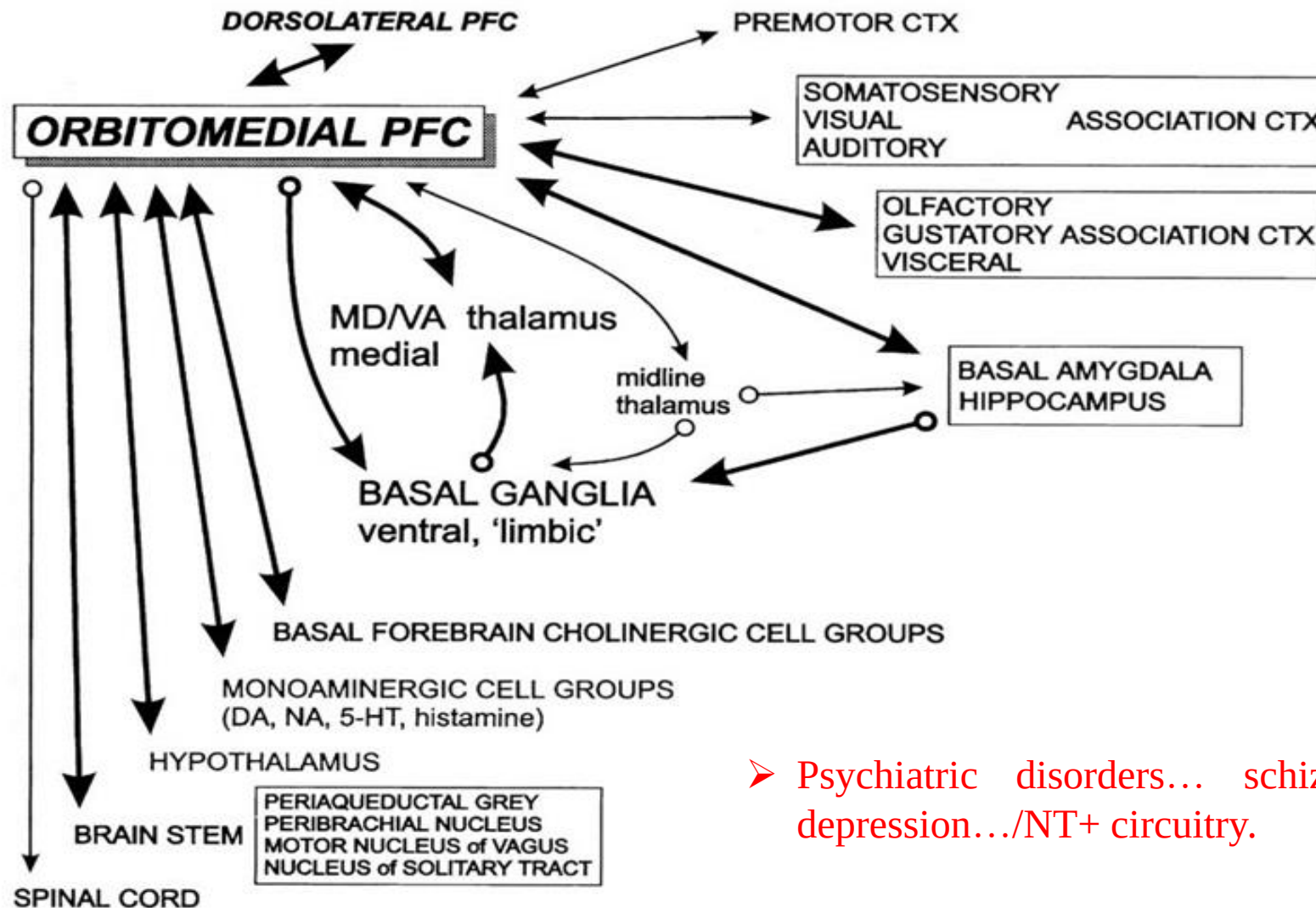


Motor Association Area

2. Orbitomedial Prefrontal Cortex

- Interconnected with amygdala + hypothalamus + nucleus accumbens + brainstem nuclei.
- Involved in **arousal** (NA + Cholinergic) and are responsible for the regulation of **emotions...** (*motivational states...*)
- Prefrontal cortex → premotor + primary motor areas + basal ganglia → encode for the motor expression of emotions and behavior
- **NB** → **Play a critical role in executive control of behavior**

Connections of Orbitomedial Prefrontal Cortex



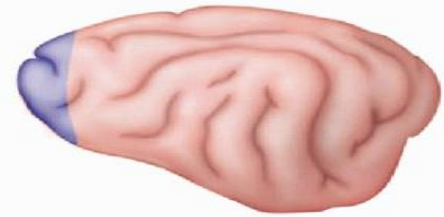
➤ Psychiatric disorders... schizophrenia ... depression.../NT+ circuitry.

Prefrontal cortex

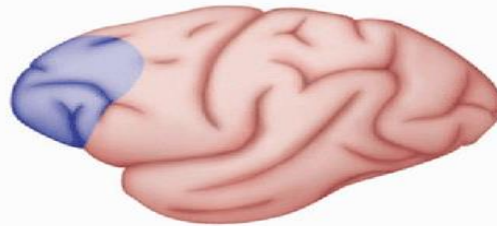
Size:



Squirrel monkey



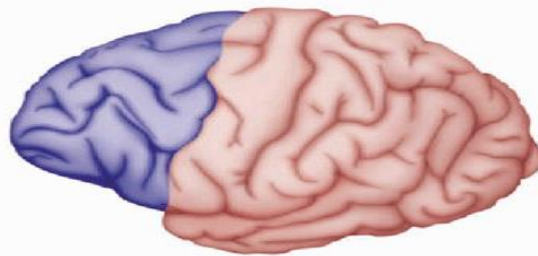
Cat



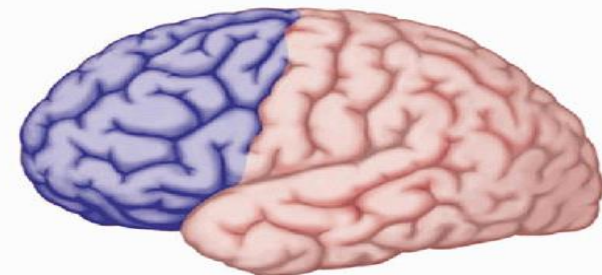
Rhesus monkey



Dog



Chimpanzee

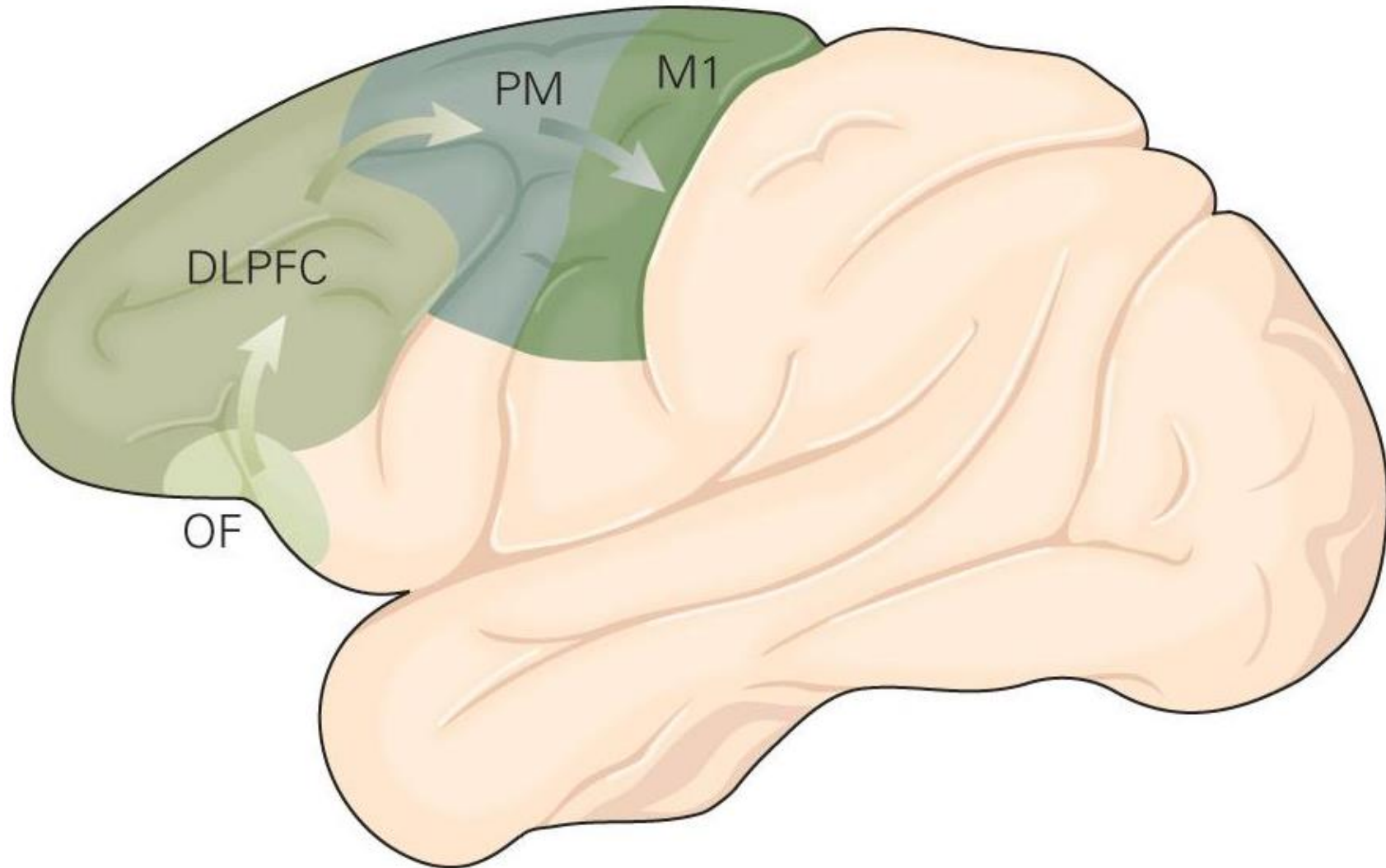


Human

Motor Association Area

- Regions of the frontal lobe are connected in series:
 - Orbitofrontal-ventromedial prefrontal cortex/OF → dorsolateral prefrontal cortex/DLPFC → premotor cortex/PM → primary motor cortex/M1 → spinal motor nerves
- Emotional + cognitive processes
 - Cognitive control of motor behavior
 - Emotional processes associated with the executive control of behavior

Frontal Cortex Connections



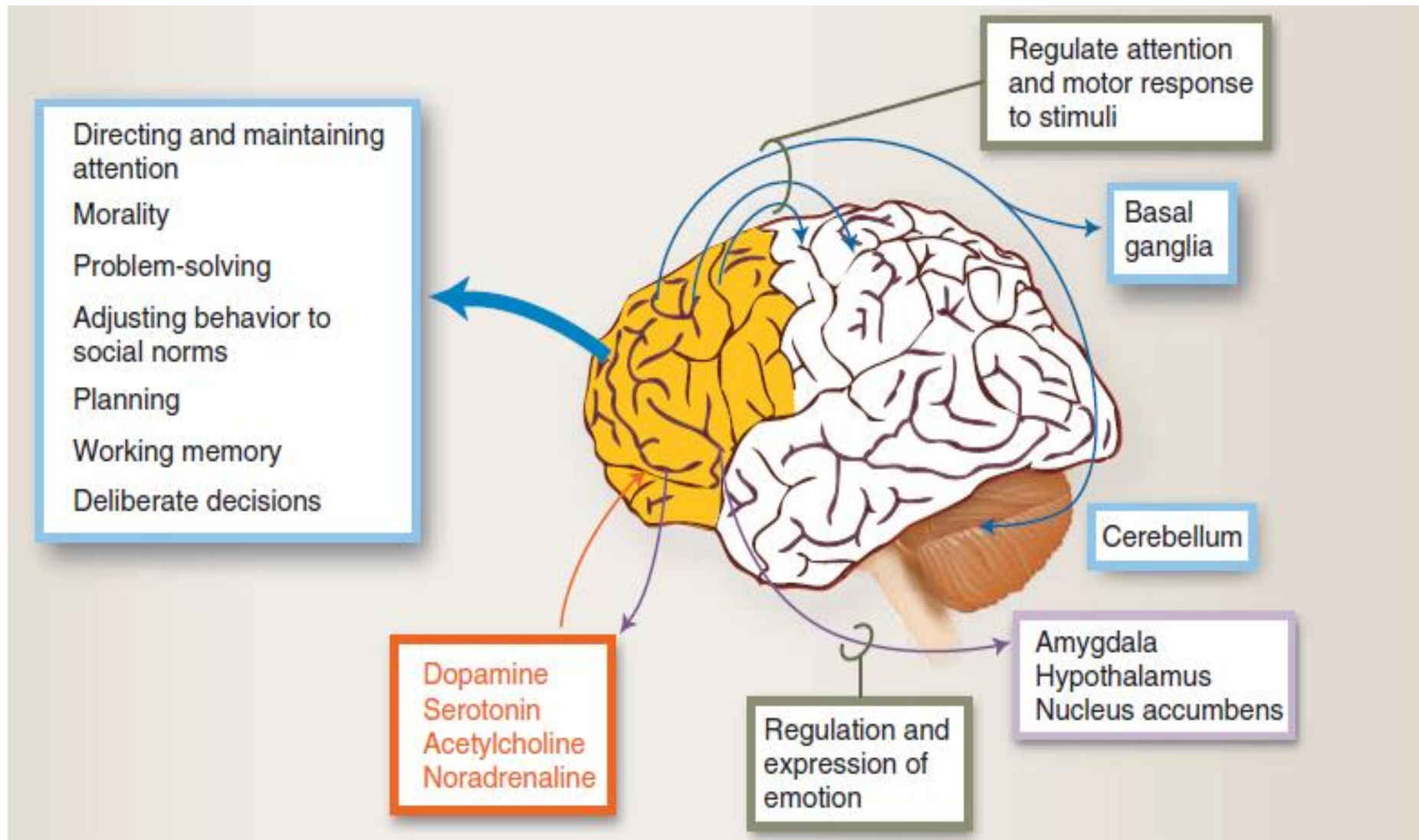
Motor Association Area

- Functions of Frontal Association Area:
 - Important for **higher order processing** + integrating + interpreting motor information and activity
 - Play a role in **anticipating or ‘planning’** a voluntary movement
 - Lesion in motor association area abolishes ***creativity*** + ***planning***
 - It gives us our **‘personality’** and lets us adjust our behavior to moral and social norms

Motor Association Area

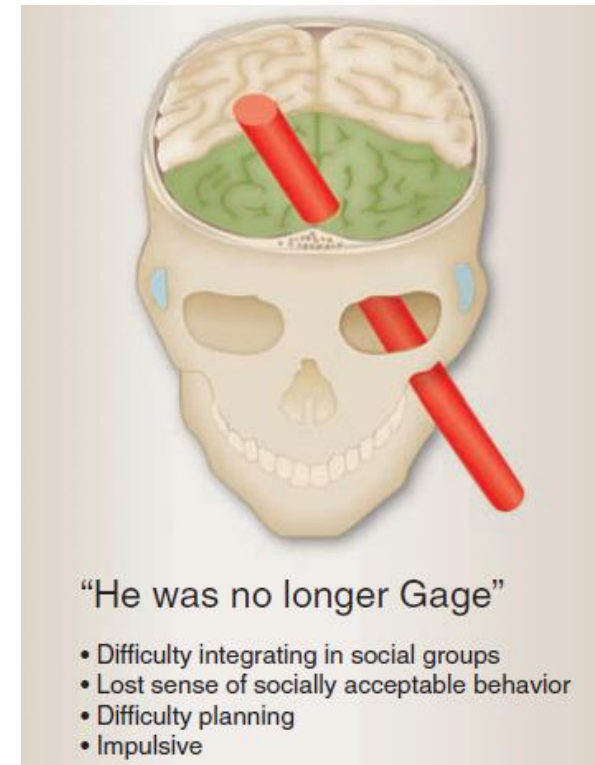
- Functions of Frontal Association Area:
 - *Executive function* → play a role in planning + problem solving as well as directing + maintaining attention on a particular situation or task
 - *Working memory* + our morality + thought + judgment + attention
 - Prioritizing and focusing on relevant themes rather than irrelevant details
 - Allows us to make deliberate decision about our behavior and adapt it to specific situations... (*Goal-directed motor behavior...*)

Prefrontal cortex: *The executive brain!*



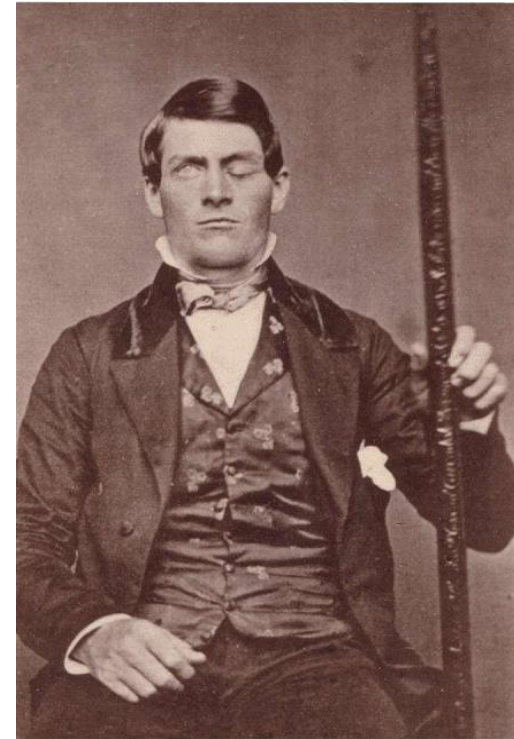
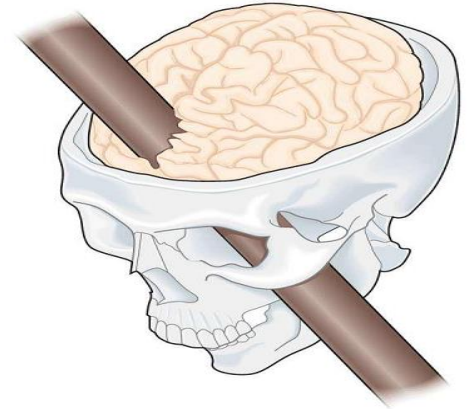
Motor Association Area

- Frontal Association Area:
 - Monoaminergic systems of the brainstem (high density of DA, NA, 5-HT receptors) + Ach
 - Psychiatric disorders → schizophrenia, depression.../NT+ circuitry
- Lesions:
 - Change in personality + loss of ability
 - Phineas Gage



Motor Association Area

- Prefrontal cortex and Phineas Gage...
 - Change in the personality + loss of ability
 - Disorganized behavior, disinhibition
 - Lack of concern for the future
 - Consistently poor decision-making
 - Impulsiveness
 - Failure to obey rules/antisocial
 - Lack of social graces
 - Disposed to imitation
 - Emotional lability



Phineas Gage...1848